

Adv SQL → next session

→ Transactions

→ Practical

Agenda :

→ Aggregate Fn. → 5 types

↳ GROUP BY

↳ HAVING clause

→ Inbuilt Fnc → NUMERIC
STRINGS
→ DATE

→ Doubts

Aggregate → combine, grouping things

Till now.

{ SELECT * FROM < >
WHERE
} Row by row operations

Problem → find ^{avg} max psp from a students
table

↓
problem solving percentage

Aim : Combine rows and perform some
operations on that group.

Aggregate Fn

fn (set of values)



single value

$$\text{sum} (1, 2, 3, 7) \Rightarrow 13$$

$$\text{Max} (1, 2, 3, 7) \Rightarrow 7$$

$$\text{Min} (1, 2, 2, 7) \Rightarrow 1$$

$$\text{count} (1, 2, 2, 7) \Rightarrow 4$$

$$\text{Avg} (1, 2, 3, 7) \Rightarrow 3.25 \left(\frac{13}{4} \right)$$

What happens if we have NULL values passed to the aggregate fn.

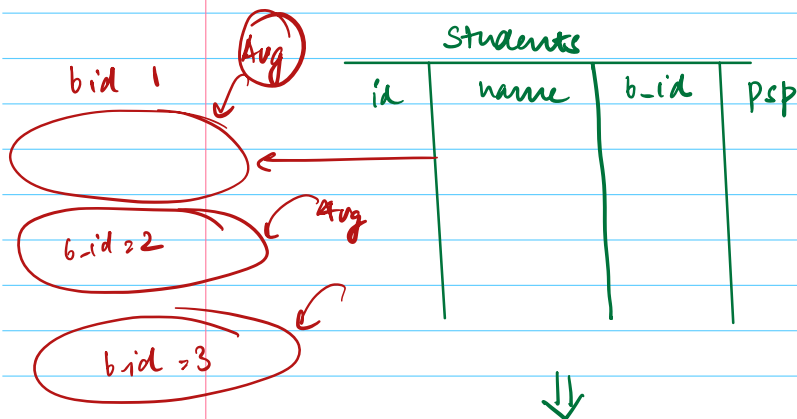
$$\text{COUNT} (1, 3, 7, \text{NULL}, 8, \text{NULL})$$



④ ✓

Aggregate fns ignore the NULL values

Till now, all aggregate fun work on the full table



Find avg
psp for every
batch?

```
SELECT AVG(psp)
FROM students
WHERE b-id = 1;
```

Avg psp of
students in batch 1

⇓

b-id	avg(psp)
1	?
2	?
3	?
4	?

GROUP BY

STEPS :

① Group different rows of the table

b-id = 1

b-id = 2

② Perform an operation on these groups
to get values

```

SELECT b_id, avg(psp)
FROM Students
GROUP BY b_id;

```

Q Find batches where students have avg psp > 80

where clause?

✓
 SELECT b_id, avg(psp)
 FROM Students
 GROUP BY b_id
WHERE avg(psp) > 80
 COMPILE ERROR

only works on the cols of original table

SELECT b_id, avg(psp)
 FROM Students
 WHERE psp > 80
 group by b_id

Will work
 ↓
 but output
 will be
 incorrect

Filtering students with
 psp > 80
 ↓
 and then
 group by.

Students				
id	name	b_id	ps	
1	A	1	81	→ Gp-1
2	B	1	79	
3	C	2	82	→ Gp2
4	D	2	75	
5	E	2	85	

Gp-1
 1 0
 2 2
 2 2
 Gp2
 avg psp of batch
 of students psp > 80

HAVING clause

- WHERE can't be used with group by.
- Filter after the group by → HAVING

```
[ SELECT b_id, avg(psp)
  FROM students
  GROUP BY b_id
  HAVING avg(psp) > 80 ]
```

WHERE

- Filter out rows
- before group by
- original cols of the table to filter

HAVING

- Filter out groups
- after group by
- cols created after group by

- ① FROM
- ② WHERE
- ③ GROUP BY
- ④ HAVING.



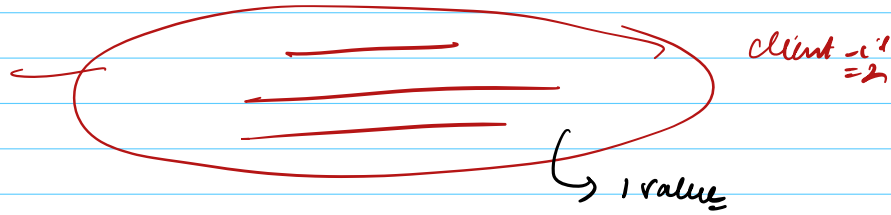
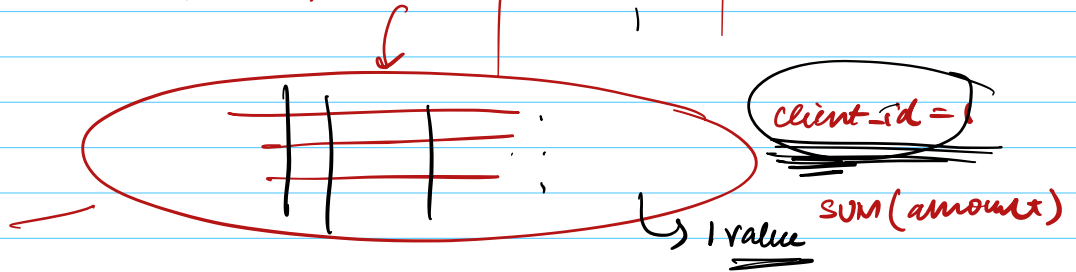
SQL

FROM

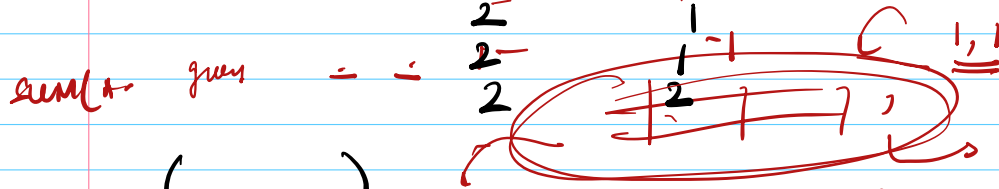
GROUP BY

INVOICES

id	date	client_id	amount	payment
----	------	-----------	--------	---------



id	client_id	payment_method	amount
----	-----------	----------------	--------



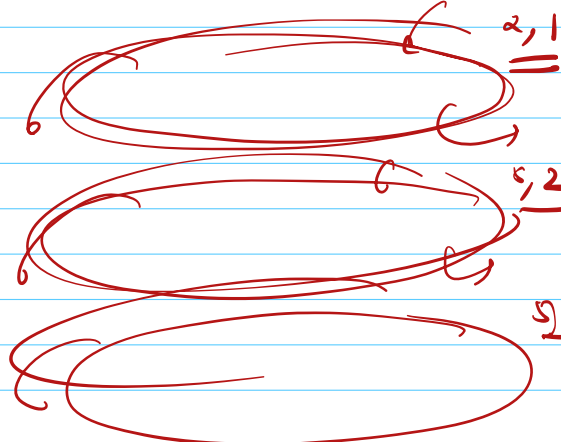
(= =)

(1, 1) →

(1, 2) →

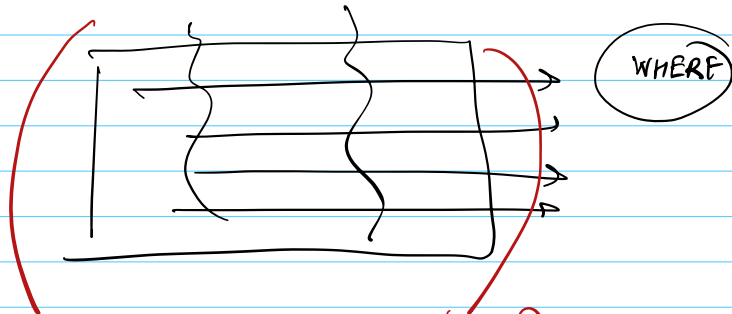
(2, 1) →

(2, 2) →



10 mins break

10:20 PM
2



Agg → (Set) → 1 single

GROUP BY

FORM groups based on columns specified

col1 col2

(x, y)

Every distinct pair
of (x, y)

↓
separate group

